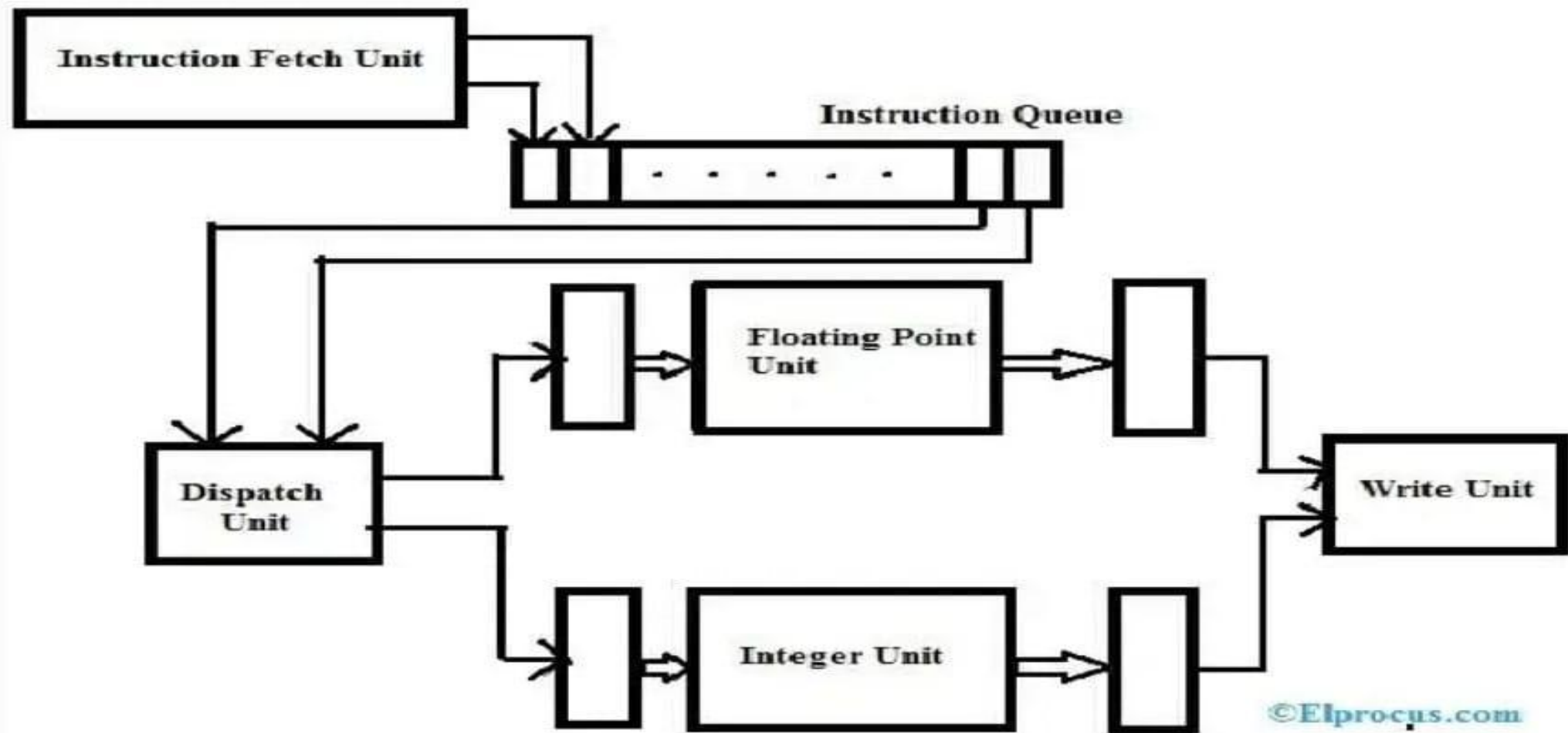


Superscalar Architecture

Definition

- A **superscalar processor** (or **multiple-issue processor**) is a CPU that implements a form of parallelism called instruction-level parallelism within a single processor.
- A superscalar processor can execute more than one instruction during a clock cycle by simultaneously dispatching multiple instructions to different execution units on the processor.

Superscalar Processor Architecture



1. Integer Unit (ALU - Arithmetic Logic Unit)

The Integer Unit, also known as the Arithmetic Logic Unit (ALU), performs arithmetic and logical operations on integer data,

such as:- Addition and subtraction

- Multiplication and division
- Bitwise operations (AND, OR, XOR)
- Comparisons (equal, greater than, less than)

The Integer Unit is responsible for executing instructions that involve integer arithmetic, logical operations, and control flow.

2. Floating-Point Unit (FPU)

The Floating-Point Unit (FPU) is a specialized execution unit designed to handle floating-point operations, such as:- Floating-point addition and subtraction

- Floating-point multiplication and division
- Trigonometric functions (sin, cos, tan)
- Exponential and logarithmic functions

The FPU is optimized for floating-point calculations, which are commonly used in scientific simulations, graphics rendering, and other applications that require high-precision arithmetic.

Advantages

- **Increased Instruction Throughput:** Its ability to execute multiple instructions concurrently, often in a single clock cycle. This results in a substantial increase in instruction throughput compared to scalar processors, which execute one instruction at a time.
- **Improved Performance:** Superscalar processors excel in handling a wide range of tasks, from basic arithmetic operations to complex calculations and data manipulations.

- **Efficient Resource Utilization:** Superscalar processors have multiple specialized execution units that handle different types of instructions simultaneously, allowing the CPU to efficiently use hardware resources and improve performance.
- **Parallelism Exploitation:** Superscalar architectures leverage instruction-level parallelism (ILP), which enables the concurrent execution of independent instructions. The CPU's instruction scheduler identifies and dispatches these independent instructions to different execution units simultaneously. This parallelism significantly reduces the time needed to complete a task and maximizes CPU utilization.
- **Out-of-Order Execution:** Instructions are executed as operands become available, rather than in strict program order, helping to avoid stalls.

Disadvantages

- **Increased Complexity:** Designing and implementing superscalar processors is more complicated due to the need for advanced scheduling and dependency management.
- **Higher Power Consumption:** They typically consume more power because of multiple execution units and additional control logic.
- **Instruction Dependencies:** Superscalar processors can still face pipeline stalls due to instruction dependencies, where one instruction must wait for the result of a previous one before continuing.

CONCLUSION

- Superscalar architectures boost CPU throughput by executing multiple instructions per clock cycle, increasing performance without raising the clock rate.
- All general-purpose CPUs developed since about 1998 are superscalar.
- The major problem of executing multiple instructions in a scalar program is the handling of data dependencies. If data dependencies are not effectively handled, it is difficult to achieve an execution rate of more than one instruction per clock cycle.